#	Op #	Operation Description	Equipment / Tools	Key Control Points (KCP)
	10	**PCB Inspection & Prep** **Remove mainboard from ESD shielding bag. Inspect PCB for scratches, bent pins, or missing components. Place on ESD mat.**	ESD Strap Visual Magnifier (x10)	**ESD Check: Strap resistance < 10MΩ** **Visual: No bent pins > 5 degrees**
	20	**Apply Thermal Interface Material** **Dispense 0.5g of thermal compound (Arctic Silver) onto the CPU heat spreader. Use stencil S-20 to ensure a 0.2mm uniform layer.**	Auto-Dispenser D-500 Stencil S-20	**Weight: 0.50g ± 0.05g** **Coverage: 100% of die area covered** **Thickness: 0.2mm ± 0.02mm**
	30	**Install Heatsink** **Align heatsink guide pins with PCB holes. Press down firmly until flush. Secure with 4x M3 screws in a cross pattern (1-3-2-4).**	Torque Driver (Preset 1.2Nm)	**Torque: 1.2Nm ± 0.1Nm** **Visual: Heatsink flush with board** **Gap Check: < 0.1mm gap**
	40	**Connect Fan Harness** **Plug the 4-pin CPU fan connector into header CPU_FAN1. Ensure the locking tab engages. Route wire through clip A.**	Needle Nose Pliers	**Tactile: Audible "click" upon insertion** **Visual: Wire routing must not touch heatsink fins**
	50	**Install WiFi Module** **Insert M.2 WiFi card into slot E. Secure with 1x M2 screw. Snap antenna cables (Main & Aux) onto connectors.**	Micro-Torque Driver (0.3Nm)	**Torque: 0.3Nm ± 0.05Nm** **Connection: Antenna connectors must rotate freely**
	60	**Chassis Integration** **Place motherboard assembly into the bottom chassis. Verify I/O shield alignment. Install 6x standoff screws.**	Electric Driver (Low Speed)	**Visual: I/O ports centered in cutouts** **Hardware: All 6 screws present and seated**
	70	**Firmware Flashing** **Connect programming cable to JTAG port. Run script flash_v2.0.bat. Wait for "PASS" LED (Green).**	Programming Station P-1 Test Fixture	**Software: Verify Checksum matches release notes** **Indicator: Green LED = Success, Red = Fail**

#	Op #	Operation Description	Equipment / Tools	Key Control Points (KCP)
	80	**Functional Test** **Power on unit. Verify boot sequence on test monitor. Check fan spin and WiFi signal strength (> -60dBm).**	Power Supply (12V) Test Monitor	**Power: Current draw < 1.5A** **Functional: Boot to OS in < 15 seconds**